Conclusion:
MWC are common in the management of STS and are positively correlated with age, neoadjuvant radiotherapy, BMI and lower limb localization. To ensure other oncological treatments can be performed after surgery, a nomogram is proposed here to assess patient's individual risk of MWC. A validation cohort is currently underway to validate our model.

Keywords: wound complication, radiotherapy, sarcomas

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Analysis of predictive factors for distant metastatic recurrence following carbon-ion radiotherapy for choroidal melanoma

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Purpose/Objective:
This study aims to identify predictive factors for distant metastasis after carbon-ion radiotherapy for choroidal melanoma.

Material/Methods:
We retrospectively analyzed predictive factors of distant metastatic recurrence in patients with choroidal melanoma treated at our institution between April 2001 and March 2021. Following carbon-ion radiotherapy, whole-body CT scans, liver MRI or liver ultrasound, and orbital MRI were performed every 3 months for the first 2 years, and approximately every 6 months thereafter. Univariate analysis was conducted using the log-rank test, and multivariate analysis was performed with the Cox proportional hazards model.

Results:
A total of 273 cases were reviewed. Patient characteristics were as follows: median age of 56 years (range 15-86 years); male-to-female ratio of 130:143; median tumor diameter of 11.0 mm (range 4.0-20.0 mm); median tumor thickness of 7.5 mm (range 1.8-17.0 mm). Tumor size categories were distributed as 1:2:3:4 = 50:104:105:14. Ciliary body adjacent was observed in 29 cases (11%), and papilla adjacent in 60 cases (22%). The prescribed dose started at 60-70 Gy in 5 fractions, followed by 64 Gy in 4 fractions (biologically effective dose) and the currently used prescription of 68 Gy in 4 fractions (biologically effective dose) from September 2016.
With a median follow-up of 78 months (range 5-253 months), distant metastasis was the main form of recurrence, with cumulative incidences of 23% at 5 years and 33% at 10 years. Among the 81 patients with distant metastasis, the liver was the most common site (55 cases), followed by the lungs (17 cases). The 5- and 10-year local control rates were 95% and 92%, respectively; and 5- and 10-year overall survival rates were 86% and 76%. In univariate analysis, gender, size category, ciliary body adjacent, and papilla adjacent were significant risk factors for distant